# Semester Research Progress Report

Throughout the Spring 2026 semester, my work consisted of computation geometry, some structural analysis, and experimental apparatus design within the MIDL laboratory. The initial phase of the work focused on establishing using my previous mathematical foundation for geometric modeling, specifically regarding point and polygon location. This then progressed to voronoi generation, then mesh generation using voronoi, then creating wigner seitz cells. Over the semester, I realized how overcomplicated I made the problem to be. Besides computational geometry, I was able to get more familiar with ANSYS and FEA. Also, I was able to learn about 3D printing, prototyping and experimental setup design.

## Computational Geometry:

Building upon foundational principles, the computational work related to research started by generating polygons using points using vertices and edges with pre-determined points. It used lists of vertices and edges, then used closest edges to graph the polygon. With these polygons, tests were done using random points and distinguishing whether the random point was inside a polygon using the nearest edge. The script iterated through each edge point of the polygon to calculate the shortest perpendicular distance between the point and line segment. After identifying the closest edge point, it used a cross product between the polygon's edge and vector point from edge to the test point. If the cross product was deemed to be positive it meant it was in the counter clockwise direction. This logic was using Jordan's curve theorem where it differentiates if a point is on the inside, which essentially means the random point is to the left of the edge. The final part of the code helped visualize the polygon and random point and applied a white color to the random point if it was deemed to be inside the polygon.

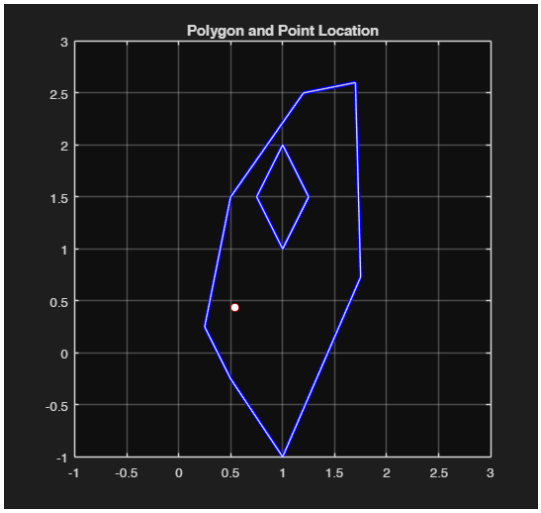



Figure 1: Point in Polygon Test

After the start of polygon generation, there was a transition of creating polygons using Voronoi. This script generated a random set of points over a domain. Then, using voronoiin, the code computed how the point's voronoi cells will be divided across the domain. This script also gave a list of vertices of the polygons and the edges of each polygon. Then plotted with each polygon colored and outlined. During this project, I realized that voronoi needs to have a boundary to make sure that the cells don't get spread to infinity. This caused each polygon to be extremely stretched and very irregular.

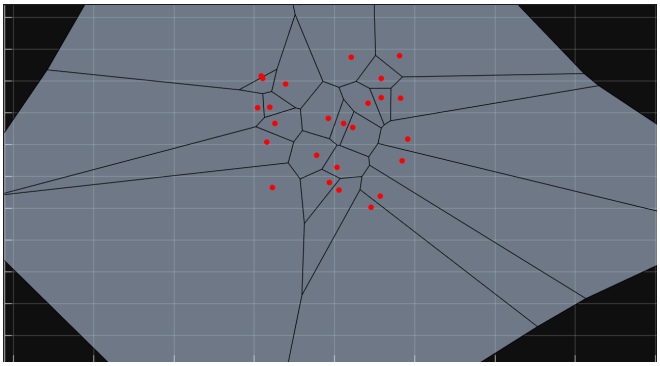


Figure 2: Voronoi generation with irregular polygons

After the initial part of voronoi generation, the focus shifted toward making the diagram behave better near boundaries and having points generated in a pattern. Firstly, the points were generated using a certain distance between each point and generated until a determined boundary. After using a pattern of points, the next problem was creating a way to have a cutoff for a boundary to not create irregular polygons. To resolve the issue, points (ghost points) near the edge of the domain were mirrored around the boundary to make polygons look more “regular” closed polygons. This just helped make edges for the points inside the domain to be shaped around.

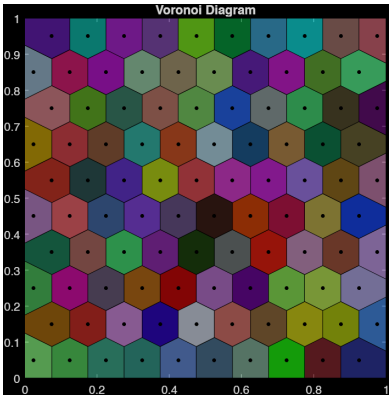
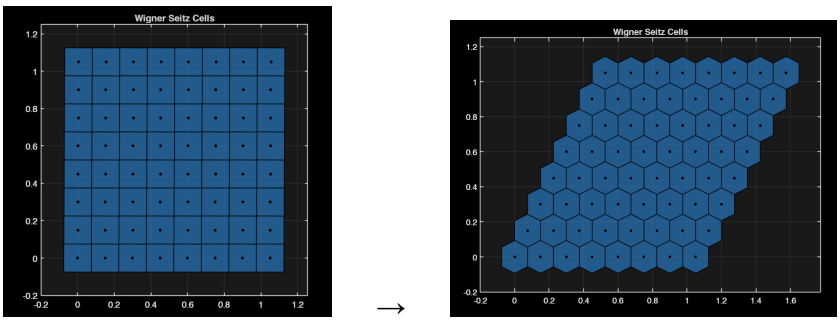



Figure 3: Patterned Voronoi

After fixing the boundary issues with voronoi polygons, the project transitioned to lattice generation using Wigner Seitz cells. This used the same approach of defining structures using simple distance and random points in patterns and was overly complicated. These points made Wigner Seitz cells were essentially Voronoi cells that were shifted using a shear amount and made into an animation to show the transformed structure.




Figures 4 & 5: Wigner Seitz cells transformation

This approach made the project only valid or applicable for a specific case and also showed how over complicated the code was. It was using manual calculations to create these structures. These manual calculations make the scripts too complex. After the review with Dr. Krishnamurthy, it was evident that I need to learn the fundamentals of Linear Algebra to make the project modular and be applicable to different polygons.

The next approach for lattice generation used basis vectors to define a 2D bravais lattice. It uses a set of rows and columns or linear combination of basis vectors to create a whole lattice. Then using linear transformation by a shear amount to the linearly independent basis vector that has a vertical component, causing a lean in the cell, while maintaining constant y values. The lattice cells are then made as Wigner Seitz cells using voronoi and then plotted on the domain and animated to show the transformation. By using basis vectors, the script looked and worked simpler.

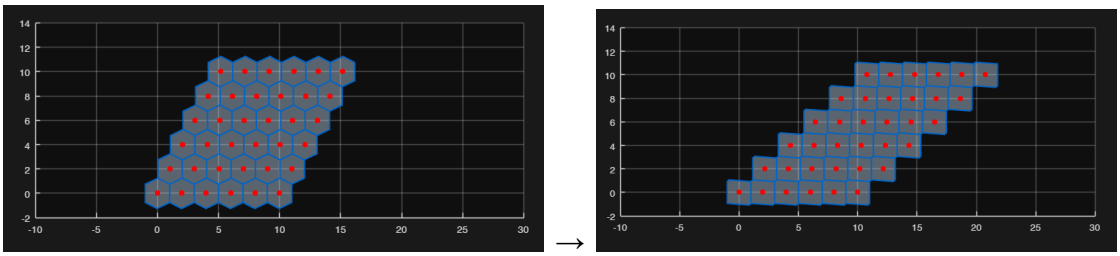
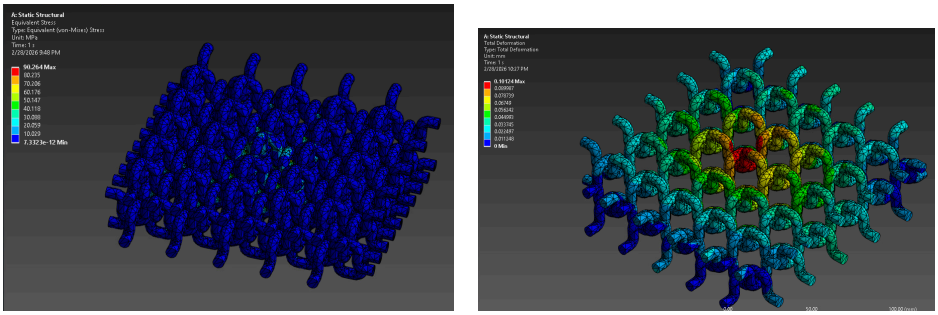


Figure 6 & 7: Wigner Seitz cell transformation using Linear Algebra fundamentals

**FEA and ANSYS Introduction:**

In addition to coding, I also learned a bit how to use ANSYS and work with Finite Element Analysis. This was at the beginning of the semester where I used one of the geometries that replicates weaves. I was able to perform static structural analysis to calculate the deformation and Von Mises Stress on the geometry. This included how to setup the model with material properties, generating mesh, and plots.



Figures 8 & 9: ANSYS analysis on custom weave geometry

**Kundt's Tube Construction:**

Finally, another component of the semester's work involved developing a model of a Kundt's tube to see the absorptivity of a 3D printed geometrically complex sample like weaves. Following a review of the "Development of a low cost impedance tube to measure acoustic absorption and transmission loss of materials" by Dr. Mohan D. Rao and Mr. Satyajeet P. Deshpande, I was able to understand the purpose of the model and create a 3D detailed CAD model of the Kundt's tube.

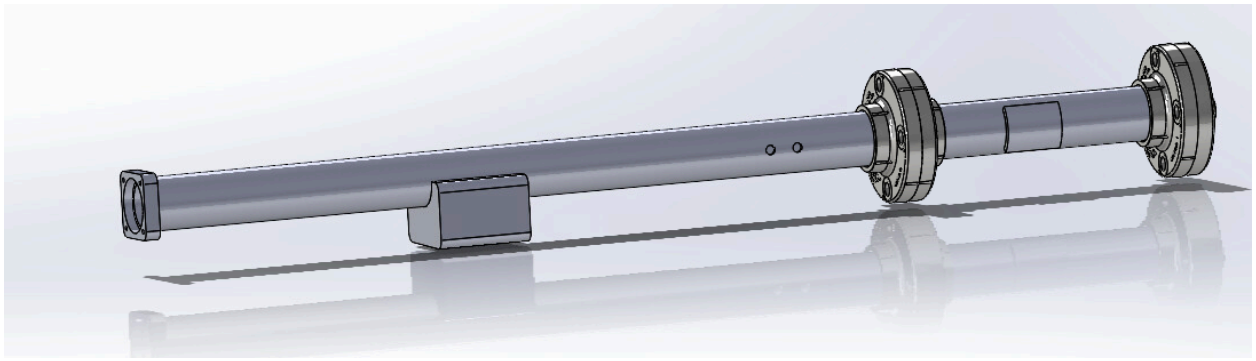


Figure 10: CAD model of Kundt's tube.

This Kundt's tube model helped me figure out how the Kundt's tube was meant to be constructed and what components I would need to 3D print. Some parts that were 3D printed were the flanges, hex bolts, washers, nuts and the reducer to connect the sound source to the model. The most tedious part of this step was creating the reducer without having the dimensions of the bolts holes. This led me to do multiple manual measurements, checking measurements of the product online, and multiple prototypes of the reducer. After creating and 3D printing every part, I was able to make a real model for the Kundt's tube.



Figure 11: Kundt's Tube Model

In conclusion, this semester I was able to learn that there are different approaches to every problem and that I should try to research more instead of sticking to the knowledge I'm comfortable with to learn better ways to solve problems. If I was able to research or at least learn the basics of Linear Algebra through videos online, I would've been able to create each script easily and with fewer lines of code. In terms of FEA, I learned the basics of ANSYS and understanding of how to set up a part. Finally with the Kundt's tube, I gained more knowledge on how to 3D print, prototyping and got more experience which will be helpful for my Capstone project.